

Solution Requirements

Date	27 June 2026
Team ID	
Project Name	Rising Waters-ML Flood Prediction System
Maximum Marks	4 Marks

• **Functional Requirements (FR):** Functional Requirements describe **what the system should do**. They define the core functionalities and services that the flood prediction system must provide to users, such as accepting weather data, processing inputs, generating flood predictions, and displaying the results.

• **Non-Functional Requirements (NFR):** Non-Functional Requirements describe **how the system should perform**. They specify quality attributes such as performance, security, reliability, scalability, and usability to ensure the system operates efficiently and provides a good user experience.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Not Applicable	Not Applicable
2	Authorization levels	Single user access with no role-based authorization.	Medium
3	External interfaces	Accept weather parameters through the Flask web interface and display prediction results to users.	High
4	Transactions processing	Process user input, perform data preprocessing, execute the XGBoost model, and generate flood prediction results in real time.	High
5	Reporting	Display the prediction outcome (Flood / No Flood) along with the predicted probability on the results page.	High
6	Business rules, etc.	Use the trained XGBoost model to analyze rainfall and weather parameters and generate	High

		predictions based on the trained dataset.	
7	Compliance to laws or regulations	Use publicly available datasets and comply with applicable data privacy and ethical AI practices.	High
8	Other	Store the trained model (floods.save) and use it for accurate predictions without retraining each time.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate flood predictions quickly after receiving user input.	Prediction generated within 2-5 seconds
2	Scalability	The application should support additional weather parameters and larger datasets in future versions.	Support future model and dataset expansion without major redesign
3	Security & Data Privacy	Protect user information and ensure prediction data is securely processed.	Secure handling of user input and data
4	Reliability & Availability	The system should consistently provide accurate predictions and remain available during usage.	95% or higher system availability during normal operation
5	Usability & Accessibility	Provide a simple, user-friendly web interface that is easy to navigate.	Users should be able to complete a prediction in 3 simple steps
6	Other	Ensure the trained machine learning model maintains high prediction accuracy.	approximately 95% or above, depending on your trained model)